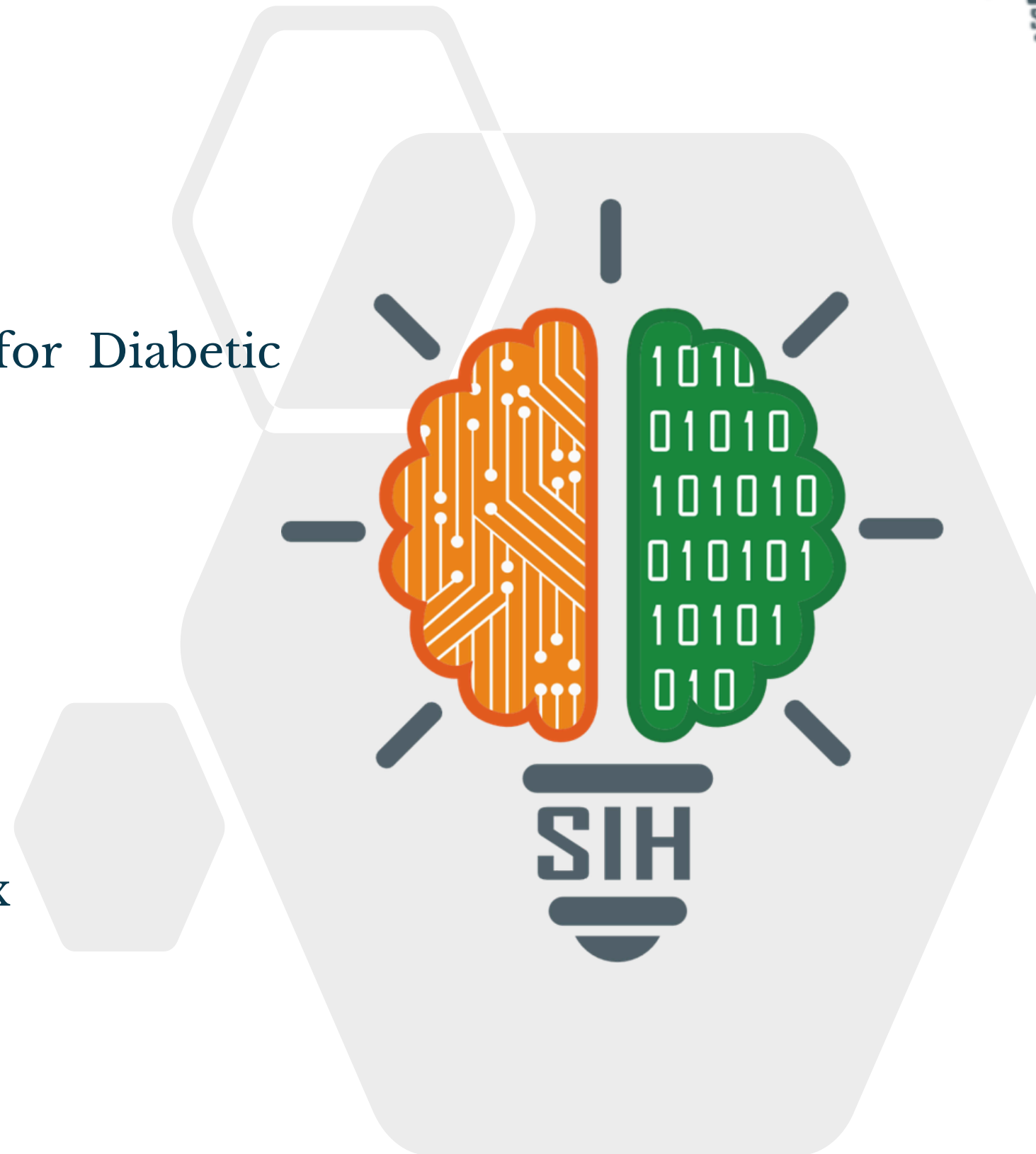


SMART INDIA HACKATHON 2026



- Problem Statement ID – SIH26038
- Problem Statement Title- Explainable AI for Diabetic Retinopathy Screening in Rural India
- Theme- MedTech / BioTech / HealthTech
- PS Category- Software
- Team ID- NSUT012
- Team Name (Registered on portal)- Veyronix



Explainable AI Diabetic Retinopathy Screening for Rural India

Kiosk-style remote screening pipeline: A MATLAB + Python pipeline that **grades diabetic retinopathy severity from a fundus photo**, **explains its reasoning** with lesion-level evidence and Grad-CAM, **routes uncertain cases to a human**, and **models district-level rollout capacity** — so a single ophthalmologist can safely screen 100,000+ patients/year through ASHA workers and PHCs.

Live Deployment: <https://drishti-frontend-c9mj.onrender.com/>

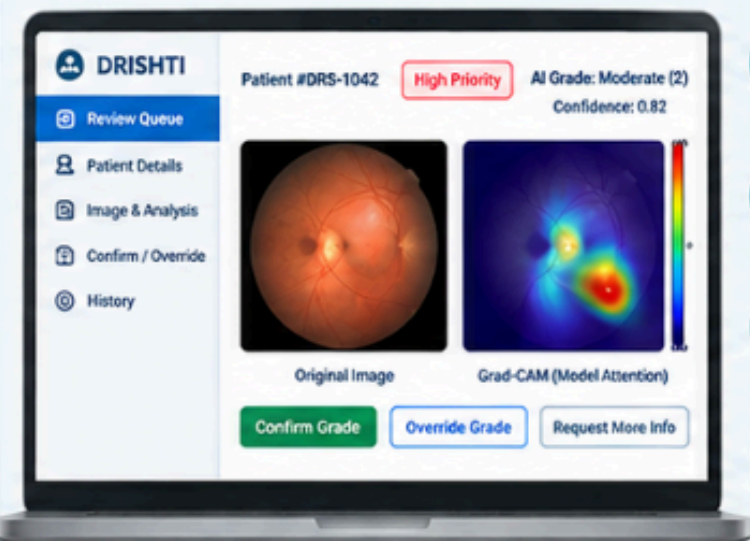
► One Backend, Three Gated Roles



ASHA / Field Worker
Android App

Capture → Screen → Guide

- Guided capture with fundus-lens adapter + offline queue
- Minimal patient registration; one-handed field workflow
- Native-language diagnosis/care audio (23 Languages (All Indian Official Languages + English))
- Works offline; syncs when connected



Ophthalmologist
Web Reviewer

Review → Verify → Confirm

- Secure login opens only the reviewer workspace and prioritised queue
- Severity + low-confidence cases first, not chronological
- Grad-CAM heatmap + lesion evidence side-by-side with the raw image
- One-click Confirm / Override updates the system-of-record grade



District Admin
Web Dashboard

Monitor → Plan → Scale

- Secure admin login opens only district analytics and operational views
- KPI header: patients screened, referral rate, pending reviews, turnaround
- Trend charts, referral funnel, per-district geographic view
- Simulink-linked staffing recommendation vs. current reviewer load

Dual-path grading
Rule-based ICDR logic cross-checks the trained model.

Human-in-the-loop safety
Calibrated confidence: gates referral to an ophthalmologist.

Validated cross-camera
Ben Graham
normalisation and
multi-source training,
tested on an untouched
external benchmark.

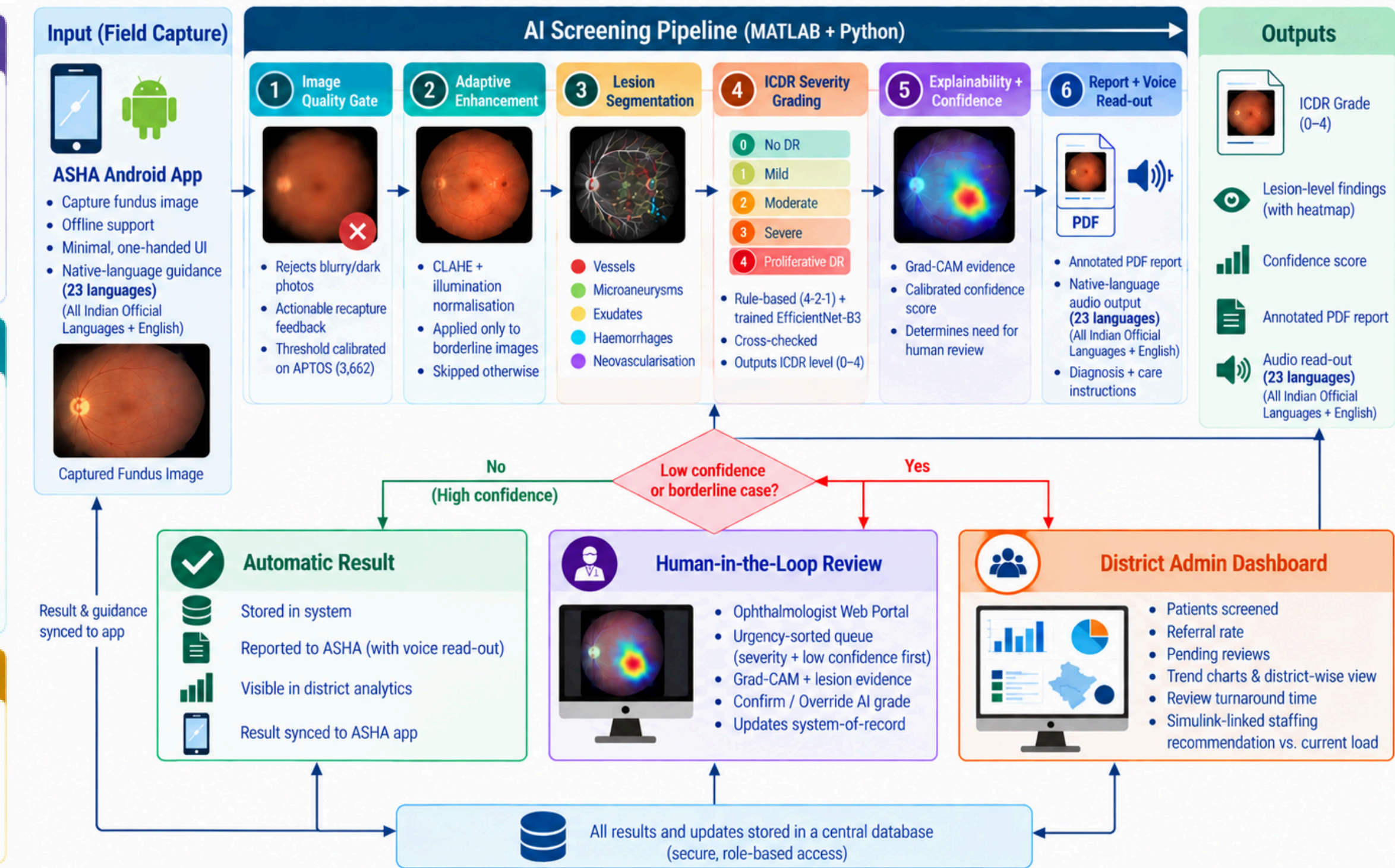
Native-language voice output
Diagnosis and care instructions read aloud in the patient's language for ASHA-led delivery.

TECHNICAL APPROACH

Technology Stack



System Architecture & Data Flow



Train fast in PyTorch → export ONNX → import into MATLAB (importNetworkFromONNX) for inference + Grad-CAM — a hybrid: fastest iteration where it matters, MathWorks tooling where the problem statement calls for it.

FEASIBILITY AND VIABILITY

Feasibility: Core classical-CV pipeline runs end-to-end today with no trained model required; the trained severity classifier is tested on Messidor-2, a benchmark completely untouched during training.

●▶▶ **Iteration Evidence (external Messidor-2 benchmark)**

Referable-DR sensitivity (target ≥ 0.90)



Final Deployed Model — Weighted Ensemble

Composition: I07 (0.376) + K10 (0.086) + K11 (0.538) — 3 independently seeded checkpoints

Sensitivity / Specificity: 0.7352 / 0.8508 • QWK: 0.5509

Effective sensitivity (AI + review): 0.8928 — near target

Review rate: 43.6% → ~73% reviewer utilization at 100k patients/yr (sustainable margin)

●▶▶ **Key Risks — Identified & Addressed Through Testing**

Risk	Why it matters	Mitigation strategy
Cross-camera domain shift	Confirmed via external Messidor-2 testing	Ben Graham color normalisation + multi-source (APTOS+IDRiD) training
Rare-class (severe/PDR) data scarcity	Causes unstable, biased predictions	Class-weighted sampling + ordinal-aware loss + seed/checkpoint selection
Silent misdiagnosis risk	Any AI-alone system can be confidently wrong	Calibrated confidence gates mandatory human review, never a silent miss
District rollout planning	Unknown staffing/bandwidth needs	Simulink capacity model sizes reviewers & bandwidth for target volume
Field-worker adoption & digital literacy	ASHA workers need a one-handed usable tool	Android deployment of a lightweight app with native language read-outs

IMPACT AND BENEFITS

1

Social

- Early detection of preventable blindness for populations with least specialist access
- Native-language voice read-out of diagnosis + instructions removes the literacy/language barrier for ASHA-led rural delivery
- Turns ASHA workers & rural PHCs into real screening points

2

Economic

- Frees scarce ophthalmologist time for genuinely referable cases only, via automatic triage
- Avoids costlier late-stage treatment and lost productivity from preventable blindness
- Simulink capacity model lets a district right-size staffing/bandwidth investment, not guess

3

Environmental

- Fewer screening-only trips to distant clinics — lower travel cost, time, and carbon footprint
- Reduces load on already overstretched public hospital infrastructure
- Deployable on existing smartphone + low-cost fundus-lens hardware, no new infrastructure

101M

people living with diabetes in India.
ICMR-INDIAB, 2023

12.5%

of diabetics develop diabetic retinopathy.
Lancet Glob. Health, 2022

~12.6M

Indians estimated to have DR nationally.
derived: 101M × 12.5%

4.0%

have vision-threatening DR. (~4M people)
SMART-India study, 2022

Datasets, Methods & Sources

Category	Key Detail	Source
Datasets	APTOS 2019 Blindness Detection	Kaggle — primary training set (~3,662 graded fundus images)
Datasets	IDRiD (Indian Diabetic Retinopathy Image Dataset)	idrid.grand-challenge.org — segmentation masks + disease grading, Indian population
Datasets	Messidor-2 images + adjudicated grades	ADCIS (images) + Kaggle google-brain/messidor2-dr-grades (Krause et al.) — external validation benchmark
Methods	ICDR Severity Scale	Wilkinson et al., “Proposed International Clinical DR and DME Disease Severity Scales,” Ophthalmology, 2003
Methods	Confidence calibration	Guo et al., “On Calibration of Modern Neural Networks,” ICML 2017
Methods	Cross-camera color normalisation	Graham, B., Kaggle Diabetic Retinopathy Detection — winning solution, 2015
Public health	101M Indians living with diabetes	Anjana et al., ICMR-INDIAB, Lancet Diabetes & Endocrinology, 2023
Public health	12.5% DR / 4.0% vision-threatening DR prevalence	SMART-India study, Lancet Global Health, 2022
Feasibility	Low-cost smartphone fundus imaging validation	Wintergerst et al., smartphone-based fundus photography review, Ophthalmology, 2020